

Analysis

An Empirical Study of US Environmental Federalism: RCRA Enforcement From 1998 to 2011[☆]Eric Sjöberg^a, Jing Xu^{b,*}^a Department of Economics, University of Utah, Salt Lake City, USA^b School of Public Finance and Taxation, Southwestern University of Finance and Economics, Chengdu, China

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ABSTRACT

State enforcement of federal environmental legislation has long been associated with the fear of environmental freeriding and states being too lenient, potentially leading to a race-to-the bottom in enforcement. This paper analyzes the effect of enforcement decentralization from 1998 to 2011 on a range of enforcement outcomes across US states as they have been authorized to implement their own programs aligning with the Resource Conservation and Recovery Act (RCRA). In this study, we look at the association between enforcement decentralization and corresponding outcomes, namely the number of evaluations, number of detected violations, and the amount of monetary penalties. Results showed that enforcement behaviors have not changed significantly as states have assumed more enforcement responsibilities; hence, we found no evidence of a race-to-the-bottom scenario associated with RCRA decentralization.

1. Introduction

In theory, environmental regulation can address market failures and correct externalities. In practice, however, how should environmental regulation be enforced to ensure efficient outcomes? Both state and federal agencies can enforce environmental regulation. The question of which of the two types of agencies would enforce environment regulation better has been debated in the theoretical and empirical literature as well as across disciplines such as economics, sociology, and political science. In this paper, we aim to study different enforcement outcomes related to the Resource Conservation and Recovery Act (RCRA) across US states as enforcement responsibility has been transferred gradually from the US Environmental Protection Agency (EPA) to state agencies.

The benefits and drawbacks of federal and state enforcement can be summarized as follows.¹ Federal (or centralized) enforcement of environmental legislation is assumed to be just with few systematic differences in stringency across regions. Centralized enforcement also accounts for cross-jurisdictional costs, such as when lax enforcement in one region leads to poor environmental quality in neighboring regions. One disadvantage is that centralized enforcement is thought to respond inappropriately to local stringency preferences. Another potential problem is that federal agencies lack

essential knowledge about local institutions, local businesses, and the local environment and thus cannot efficiently conduct inspections and other key tasks (see Carlsson et al., 2011). State (or decentralized) enforcement, on the other hand, can adapt to local needs and preferences but does not necessarily take into account cross-jurisdictional externalities. Moreover, decentralized enforcement may be susceptible to undue influence from the local community given the camaraderie between local inspectors and industry, leading inspectors to shirk their responsibilities (e.g., Ringquist, 1993; Aimer and Goeschl, 2010; Jung and Makowsky, 2014). These considerations suggest that decentralization may result in a race-to-the bottom in environmental enforcement where states compete in clemency and leniency in an effort to appear business-friendly. Another potential downside of decentralized enforcement is unpredictable stringency across regions. Potential causes of such variations are that states' understanding of environmental rules may not be uniform, and compliance costs differ across regions (Becker, 2011b). Further, states may have different preferences regarding environmental quality (Sigman, 2003; Epstein, 2017) along with distinct coordination practices with federal authorities (Opp, 2011). The EPA has also acknowledged that federal oversight of state enforcement is poor (EPA, 2011). Taken together, the literature on the possible effects of decentralized enforcement points to substantial uncertainty

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¹ See Millimet (2014) for a more detailed discussion of this problem, or Young (2002), where the problem is discussed more generally under the apt terms *problem of scale* and *problems of interplay*.

surrounding the net effects.²

While expansive research exists on the relationship between federal enforcement and environmental outcomes (e.g., List and Gerking, 2000; Lim, 2016), few papers have focused on possible mechanisms, such as inspection frequency, punitive actions, and so on to examine the extent to which federal and state enforcement differ in their respective effects on the environment. In the present empirical analysis, we study the association between enforcement outcomes and decentralization of authority from the EPA to US states related to the RCRA. The RCRA serves as a legal framework for managing hazardous and non-hazardous solid waste. We use the fact that US states can apply to the EPA to request authority to implement regulation corresponding to federal programs under the RCRA. This process, which transfers enforcement responsibility from the EPA to states, is called *authorization* and will be explained in detail in Section 2. Essentially, the more programs over which a state has authority, the more decentralized the enforcement is in that state. While the authorization process is reversible in theory, no examples exist of states voluntarily relinquishing authority or authority being revoked by the EPA for RCRA-related legislation. Unlike much of the previous literature, we do not consider decentralization a binary variable where enforcement is either centralized or decentralized (see Grooms, 2015). Instead, we construct a continuous indicator to measure the degree of decentralization for each state as enforcement responsibility is gradually transferred to the states. Our analysis focuses on three broad dimensions of enforcement: enforcement actions, represented by inspection frequency; enforcement outcomes, including any identified environmental violations and corresponding proposed punishments (i.e., monetary fines); and the violation/inspection ratio, used to indicate efficiency of environmental enforcement.

We found that as authorization increases over time, there is no statistically measurable effect on enforcement behavior. These results were robust for all outcomes (i.e., inspections, violations, penalties, and efficiency measures). The voluntary and gradual delegation of enforcement to state agencies thus seems to have no explicit adverse effects on enforcement outcomes, suggesting there is no difference between federal and state agencies' social-ecological system fit as defined in Epstein et al. (2015), at least not with regard to enforcement. It should be noted, however, that freeriding in terms of environmental outcomes has been associated with decentralized enforcement (e.g., Sigman, 2005; Helland, 1998) and for certain enforcement outcomes, as noted in Grooms (2015). We discuss potential reasons for these divergent results in Section 4.

Environmental federalism has been addressed at length in earlier literature reviews. Millimet (2014) and Dijkstra and Fredriksson (2010) discussed the effects of decentralization on enforcement behavior. Laskowski et al. (2005) reviewed environmental federalism in the US. Theoretical scholarship on environmental federalism is extensive. Proposed determinants of an optimal enforcement level include the concavity or convexity of the marginal abatement cost function (Banzhaf and Chupp, 2012), the tradeoff between how well the EPA internalizes externalities against the enforcement cost advantage of local authorities (D'Amato and Valentini, 2011), and state environmental preferences (Chang et al., 2014).

Extant empirical literature on environmental enforcement can be divided into two broad strands based on methodology: papers using Reagan's new federalism to study the effects of a change in

enforcement, whose results exhibit a remarkable degree of homogeneity; and papers using other quasi-experimental methods, the results of which are more mixed. The main problem in estimating the effect of decentralization on enforcement outcomes is that the decision to centralize or decentralize enforcement is endogenous. To overcome this, several scholars have utilized the delegation of environmental regulation under President Reagan in the 1980s as an exogenous change³ in enforcement levels. Reagan's new federalism implied that states shouldered much of the responsibility for environmental enforcement. In practice, Reagan cut the EPA's workforce substantially, and a large portion of enforcement responsibilities was transferred to the states (Millimet, 2003).⁴ Papers using Reagan's decentralization have applied somewhat different econometric methods, but none has revealed any effects of increased decentralization on environmental outcomes (e.g., List and Gerking, 2000; Millimet, 2003; Millimet and List, 2003; Fomby and Lin, 2006). However, Millimet (2003) and Fomby and Lin (2006) each found an increase in pollution control expenditures after decentralization, which can be interpreted as increased enforcement activity.

The second strand of empirical literature uses other quasi-experiments to determine the effect of decentralization on enforcement and environmental outcomes. Here, the results are less consistent. Sigman (2005) found evidence of freeriding by states authorized under the Clean Water Act (CWA), an effect that did not exist for non-authorized states. Also, Helland (1998) discovered that states were responsive to local interests and modified enforcement behavior according to local preferences. Konisky and Woods (2010, 2012), however, did not find any differences in enforcement stringency across states under the CWA or the Clean Air Act (CAA). These findings differ from those of Woods (2006), who did find that when states were more stringent in enforcement compared to their neighbors, the more stringent states gradually became more lenient in enforcing the Surface Mining Control and Reclamation Act. Further evidence of local influence on enforcement was provided by Innes and Mitra (2015) who showed that new Republican representatives in Congress were associated with decreased inspection rates under the CAA. Finally, Grooms (2015) found that states that were more corrupt reported higher compliance rates but did not change their inspection behavior after first being authorized to enforce the CWA.

All in all, the literature found mixed evidence of whether states compete in enforcement. However, it appears that a link between a state having authority over enforcement and modifying its enforcement stringency to suit local preferences cannot be ruled out. It should be noted that most of the papers surveyed focused on either the CWA or the CAA. In this paper, we discuss similarities and differences in enforcement across environmental acts in the concluding section. The remainder of the paper is organized as follows: Section 2 presents our empirical model and data, Section 3 outlines our estimated effects of enforcement decentralization of the RCRA program in US states, and we conclude the paper with a discussion and suggestions for further research.

2. Empirical Model and Data

We aim to estimate the effect of decentralizing enforcement responsibility on enforcement behavior. Let y_i be enforcement behavior in state i . We are interested in the relationship described by

$$y_i = f(a_i, X_i)$$

where a_i is a measure of decentralization, with a higher value indicating a more decentralized state; and X_i is a vector of other factors that affect enforcement behavior, such as state industrial structure and local

² The general argument over the tradeoff between centralized and decentralized enforcement is not limited to environmental enforcement but is valid for any public good. See, for example, Ostrom et al. (1973) for a discussion on consolidating urban police agencies. Further, there is a long tradition in the public good literature of moving away from a binary view on agents as either egoistic or altruistic toward a deeper commitment to studying what types of contextual preferences can make the greatest contribution to our understanding of the problem (Dietz, 2005). We agree with this tradition and do not intend for our results to be considered analogous to the general preferences of different agencies.

³ Reagan's decentralization was somewhat unexpected due to the brief period over which the change took place (Millimet, 2003).

⁴ Interested readers can see Ozyim and Jarrell (2015) for changes under later administrations.

preferences. We can partition X_i in a vector that contains observable elements, s_i , and in a vector that consists of unobservable factors, u_i . We can control for observable factors. However, unobservable factors cause estimation issues when they are correlated with the level of decentralization, which is a choice variable for each state. For example, states with local preferences for lenient enforcement might choose a high level of decentralization, suggesting a greater degree of control over the enforcement process. This relationship implies a likely correlation between decentralization and unobserved factors; hence, a simple cross-sectional regression analysis will fail to take this into account and bias the estimated effect of authorization. It is therefore necessary to identify a plausible exogenous source of variation in authorization to derive more than a simple association between variables.

Unfortunately, exogenous variation in decentralization is not easily isolated. As described earlier, several papers have used the Reagan years as a source of exogenous variation in environmental regulation decentralization. An experiment that exogenously assigns different levels of decentralization across states is unlikely. An instrumental variable approach could solve this problem, but we have not been able to identify instruments that fulfill the relevance and exogeneity assumptions. Our identification strategy to measure the association between decentralization and enforcement was as follows: we controlled for a proxy of state environmental preferences as well as state industrial structure and then utilized variation in enforcement decentralization over time within states using a state panel of data between the years 1998 and 2011. Specifically, the variation in enforcement decentralization is represented by authorization from the EPA to states of RCRA rules,⁵ a process that is explained in detail in the data sub-section. By utilizing variation within states over time, the unobservable factors that are roughly constant over time can be controlled for by introducing state fixed effects, F_i . The unobserved factors that tend to change linearly can be controlled for by using state-specific time trends F_{it} (e.g., due to changing local preferences or state-specific gradual investments in enforcement infrastructure) where t denotes the time. The fixed effects and state-specific time trends were our main tools to address the endogeneity problem of authorization, a method not uncommon in the environmental enforcement literature (e.g., Lim, 2016; List and Gerking, 2000). While not as ideal as a randomized controlled experiment, we believe that, in combination with our explicit control variables on state environmental preferences and state industrial structure, these methods can capture the most essential factors behind authorization decisions. Further, we tested the robustness of our results over a range of specifications to uncover the sensitivity of our estimates. Our baseline regression specification is thus

$$y_{it} = \beta a_{it} + s_{it}\theta + F_i + Y_i + F_{it} + \varepsilon_{it}$$

where β and θ are parameters to be estimated. Both state and year fixed effects (F_i , Y_i) are included in our specification along with state-specific time trends. Our time and state-varying control variables are represented by the vector S_{it} . A more in-depth explanation of the different variables appears below.

2.1. Data

Our main independent variable of interest is authorization. When Congress passed the RCRA in 1976, the EPA was granted primary enforcement. A mechanism was also implemented wherein the EPA could delegate its implementation responsibilities to individual states through a rulemaking process called state authorization (Levy, 2012); the process is described thoroughly in Sigman (2003) and the RCRA Orientation Manual.⁶ A state can apply for the authority to implement the base

program and additional rules as long as the state program is at least as stringent as the federal program, a mechanism assumed in theory to prevent a potential race-to-the-bottom scenario. In practice, however, there are differences and leniency in application and enforcement across states, as noted in Innes and Mitra (2015) and Helland (1998).⁷ As is explained later in this section, an inspector has some discretion to address non-compliance as she sees fit. After the enactment of the RCRA in 1976, several states such as Delaware, Texas, and Virginia were first granted authorization of the base program in 1984. More states followed suit and applied for base program authorization and authorization for additional RCRA rules. The authorization process has continued since, with more rules being added to the RCRA and authorized to states. In 1981, nine rules could be delegated from the EPA under the RCRA; this number increased to 263 by 1998 and again to 324 by 2012. If a state has been authorized to implement a program but fails to execute it, the EPA can retract the authorization, although this has never happened to date. We gathered data from the State Authorization Tracking System (StATS) on the number of total rules for which each individual state had been authorized annually. We defined the index of authorization for state i in year t as

$$a_{it} = 100 \frac{\alpha_{it}}{r_t}, a_i \in [0, 100]$$

where α_{it} is the number of rules authorized for state i up to and including year t , and r_t is the maximum number of rules for which a state could be authorized up to and including year t . Hence, the index of authorization for state i in year t is simply the share of rules, expressed in percentage points, for which the state has been granted authority up to that point in time. We multiplied the share of rules authorized by 100 for ease of interpretation; a one-unit increase in authorization is analogous to a one percentage point (pp) increase. When a state's index grows from 0 to 100, it implies that the state has been granted more autonomous power in policy implementation. Our measure of decentralization, the authorization index, is transparent, does not vary by state, and has straightforward application; it does not involve subjective valuations of the importance of different rules. The largest discontinuity can be expected when the base program is authorized, which happened in the 1980s for most states. Due to data constraints around facility records and industry structures in the 1980s, we could not properly evaluate the effect of the initial authorization. Thus, the effect we estimated can be considered the marginal effect of additional rules once the authorization process was initiated. Because some rules may be more important than others and have a larger impact on our outcomes, assuming equal weights for each rule is a limiting assumption. However, authorization of a single rule is rare, implying that rules' importance may average out to some extent when multiple rules are authorized in the same year. The mean number of rules authorized in a year for a given state, conditional on authorization, was found to be 31.66 (median = 18), and authorization of a single rule only occurred in 5.12% of all cases. Assigning arbitrary weights to different rules was tempting, but there was no objective way to do so.

To measure enforcement outcomes, we used data on the RCRA, the principal Federal law in the U.S. governing the disposal of solid waste and hazardous waste. The enforcement details used as outcome variables were extracted from the RCRA Information System (RCRAInfo) of the Enforcement and Compliance History Online (ECHO) database. RCRAInfo contains information on hazardous waste handlers that fit any category of treatment, storage, disposal facilities, large- and small-quantity generators, and transporters. This database also documents inspection types, dates, facilities, violations, and penalty types and amounts conditional on the violation. In 2008, there were 381,628 RCRA-regulated facilities, each of which could be subjected to inspection (Opp, 2011).

⁵ Regulation rules translate the general mandate of an environmental statute into a set of requirements within which the regulated community and the agency must work.

⁶ <http://www2.epa.gov/sites/production/files/2015-07/documents/rom.pdf> (2/2/17).

⁷ See also Sjöberg (2016) for an example in a non-US setting.

Table 1Descriptive statistics for outcome and main independent variables. ΔX denotes the first difference of the variable X .

Variable	Mean	Standard deviation	Min	Max
Authorization share	65.73	25.62	0	97.79
Evaluations	426.76	401.33	2	2489
Violations	149.77	150.49	0	1263
Violations/evaluations	0.40	0.21	0	1
Fines (\$1000)	362.10	897.07	0	11,865
Δ Authorization share	2.20	9.96	–2.94	84.37
Δ Evaluations	–1.89	114.27	–746	720
Δ Violations	–1.42	44.83	–283	427
Δ Violations/evaluations	0.00	0.11	–0.68	0.87
Δ Fines (\$1000)	–22.00	1137.77	–11,865	11,865

Depending on the authorization status of a specific rule, either the EPA or a state agency initiates an inspection. There are several different kinds of inspections, such as case development inspections that are conducted as follow-up inspections when significant RCRA violations are discovered. However, the most common inspection type is a routine inspection, also called a compliance evaluation inspection (CEI). The inspection might be announced or unannounced, and unannounced inspections have been found to be stochastic and difficult for firms to predict (Stafford, 2013). If a violation is found, the inspector has several options. If the inspector considers the violation fairly minor, she has the discretion to either solve the issue on-site or advise the manager to deal with the issue and then confirm compliance during a subsequent inspection (Opp, 2011; Shimshack, 2014). For other violations, an inspector can choose to take administrative action, civil judicial action, or criminal action. The vast majority of actions are administrative, either informal or formal. An informal action usually involves a letter sent to the facility, or possibly a phone call, stating that the facility is non-compliant with a given regulation and outlining possible consequences if the issue is not corrected. An informal action can also contain a roadmap for how the facility can achieve compliance. If the necessary steps are not taken, or if the violation is significant, the EPA or the state can take formal action on the facility. A formal action consists of one of four different administrative orders that each impose enforceable legal duties: a compliance order; a corrective action order; an order to conduct monitoring, analysis, and testing; or an imminent and substantial endangerment order. Each of these can result in substantial penalties of anywhere from \$6500 to \$32,500 per day.

We used different regressands to measure enforcement across different dimensions. For a general discussion on the complexity of measuring environmental stringency, see Brunel and Levinson (2016). Our variables were denoted as follows:

- *Eval* – Number of evaluations in a state. Evaluation types were compliance assistance visits, compliance evaluation inspection, focused compliance inspection, operation and maintenance inspection, and groundwater monitoring evaluation. We omitted some evaluations that were less pertinent to states' enforcement behavior, such as case development inspections, which are only undertaken in pending enforcement cases. Our results did not change based on using all types of evaluations or the routine inspection (i.e., CEI) alone.⁸
- *Vio* – Number of violations found in a state. This number is conditional on inspections. We counted violations based on the evaluation types listed under *Eval*.
- *Fine* – The sum of all proposed monetary penalties in a state. Fines were derived from the violations counted in *vio* that resulted in

fines.

- *Vio/Eval* – The number of violations found as a share of the total number of evaluations in a state.

These four outcomes measured enforcement behavior along different dimensions. *Eval* was expected to correlate with state-provided regulatory resources, whereas *vio* and *fine* were assumed to correlate to some extent with enforcement and sanction severity. *Vio/Eval* served as a measure of stringency or efficiency which depends on inspector ability, available technology, and willingness to inspect, et cetera. For example, if this share were to fall with increasing authorization, it would indicate either that states were less stringent in their evaluations on average or that states had a poorer understanding of which operations to inspect to find violations.

Table 1 shows the descriptive statistics for the outcome variables and state authorizations, both in levels and in first differences.

Variation across time is important for any panel study. Only Alaska and Iowa exhibited no change in the authorization variable over the investigated time period. Both states' authorization index value was equal to 0 for all years, indicating that no authorization took place. All other states had some variation in the authorization index, much of which was substantial. On average, a state was authorized a set of rules roughly once every three years. Several states experienced significant changes in authorization, with the largest absolute change occurring in Virginia where the index moved from 3.04 in 1998 to 90.09 in 2011. We calculated the smallest and largest values of the authorization index for each state and then calculated the difference between these numbers. The average difference across all states (i.e., between the smallest and largest index values) was 34.77. The mean of the authorization index in 1998 was 46.07. Taken together, this trend demonstrates substantial variation in authorization shares over time. See Figs. 1 and 2 for a graphical representation of the variation in the authorization index across time and space. Fig. 1 shows the color-coded authorization shares for all states in 1998, and Fig. 2 shows authorization shares in 2011.

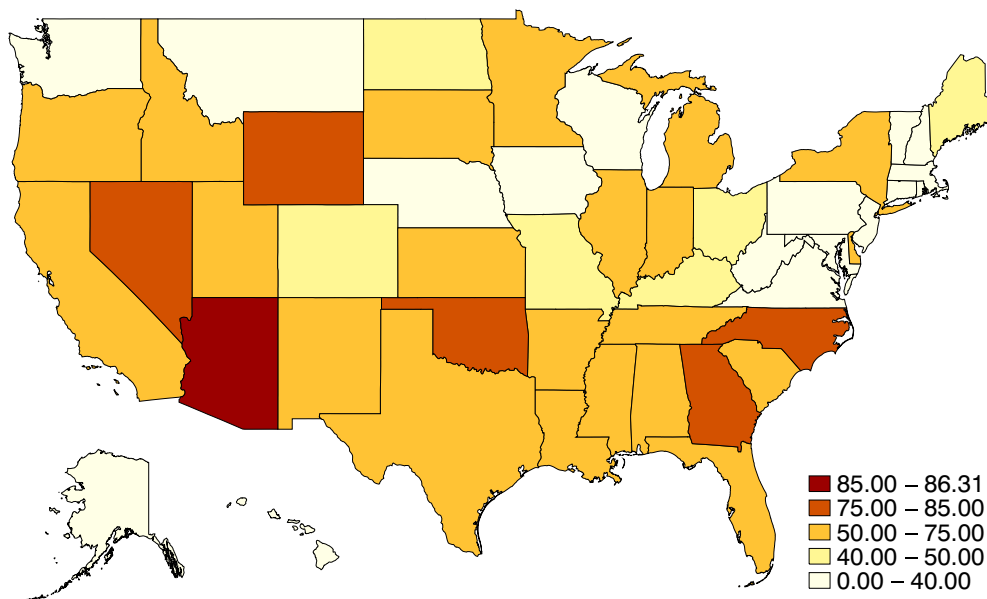
Most states' authorization shares increased over time, denoted by a darker color in Fig. 2 compared to Fig. 1. But six states—Illinois, Wyoming, Kansas, Arizona, Kentucky, and Maine—had lower authorization shares in 2011 than in 1998 due to low, but still positive, authorization rates as new rules were added to the RCRA. No state has ever been authorized on the full set of rules.

Control variables were added to the regression to capture the effects of time and state-varying factors that could possibly affect authorization and outcome variables. To account for states' environmental preferences, we followed Helland (1998) and Sigman (2005) in using League of Conservation Voters (LCV) scores. The LCV score is the proportion of time a legislator voted with the LCV's position on selected measures that are considered pro-environment. The LCV variable ranges from 0 to 100, representing the green preference of median voters in the state. The governors' attitudes towards environmental protection can also reflect and affect the state's level of environmental

⁸ For a full list of evaluations and their definitions, see the Nationally Defined Values for Evaluation Type, available at https://echo.epa.gov/system/files/ndv_eval_type_0.pdf (2/23/2017).

Authorization Shares 1998 in %

Fig. 1. Authorization shares, 1998 (in %).



Authorization Shares 2011 in %

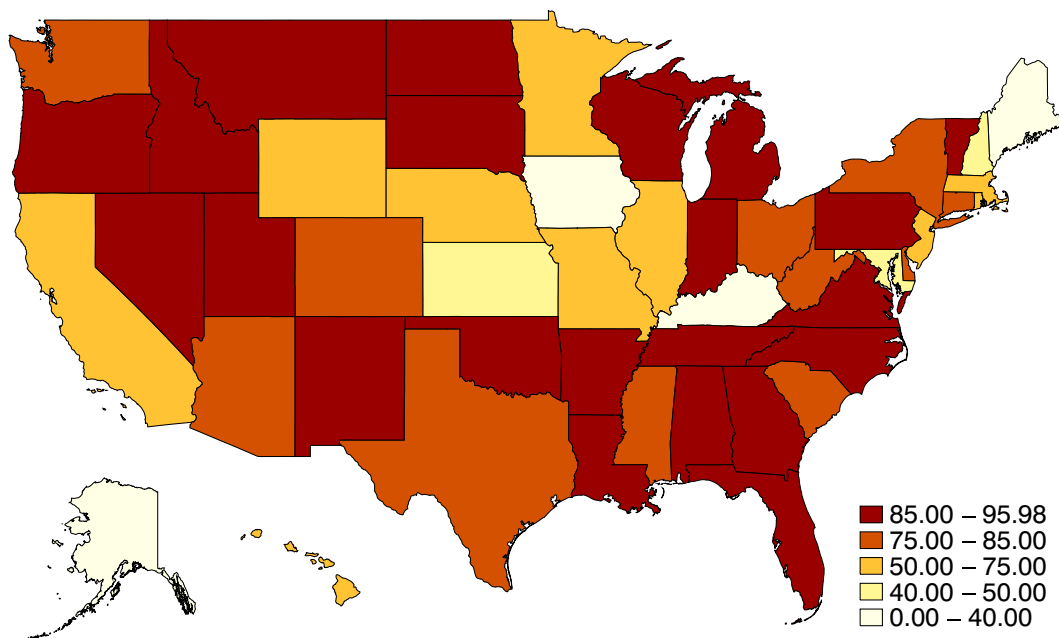


Fig. 2. Authorization shares, 2011 (in %).

concern (Boskovic, 2011). Hence, an indicator variable for party affiliation was introduced and equaled 1 if the governor was a Democratic member. We collected data on industrial structure and the number of firms from the North American Industry Classification System (NAICS). We controlled for the number of firms under NAICS codes 11 (forestry, fishing, hunting, and agricultural production), 21 (mining), 23 (construction), 31–33 (manufacturing), 48–49 (transportation and warehousing), and 56 (administrative, support, waste management, and remediation services). There were no qualitative effects of controlling for the total number of all firms. Finally, we controlled for the natural

logarithm of state GDP because economic development might affect how a state enforces environmental legislation. Standard errors were likely to be correlated within a state over time and were thus clustered on a state level across all regressions.

3. Empirical Results

We first examined how the different outcome variables and the authorization variable evolved over time in our dataset. Fig. 3 shows the evolution of the average share of authorizations across states over

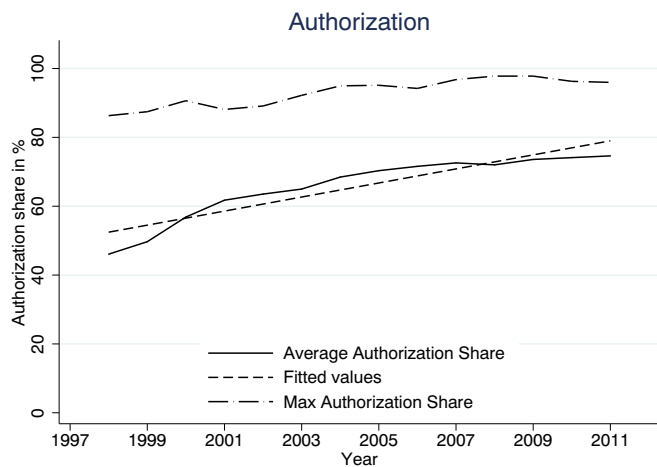


Fig. 3. Average authorization share between 1997 and 2011 with estimated trend line. Dashed-dotted line is maximum authorization share.

time as well as the maximum value; note that Alaska and Iowa demonstrated 0, the minimum value, for all years. Over time, the average value of the authorization index increased, even as new rules were added and more rules became eligible for authorization. The increasing trend has stagnated since 2006.

Figs. 4–7 show the evolution over time of this study's four different outcome variables: evaluations, violations, proposed monetary penalties, and violations as a share of evaluations. These figures also display a fitted linear time trend.

For three of our four outcome variables—average number of evaluations, average number of violations, and violations as a share of evaluation—we identified no discernable trends over the studied time period. However, the average total proposed monetary penalties declined sharply. For aggregate data, we therefore noticed an increasing trend in authorization that was not clearly correlated to any particular change in the number of evaluations or violations but was negatively correlated to the trend of proposed penalties. If related to greater enforcement decentralization, this trend could have two potential explanations: as the authorization indices increased for states, states may have still evaluated and found violations to the same extent as before

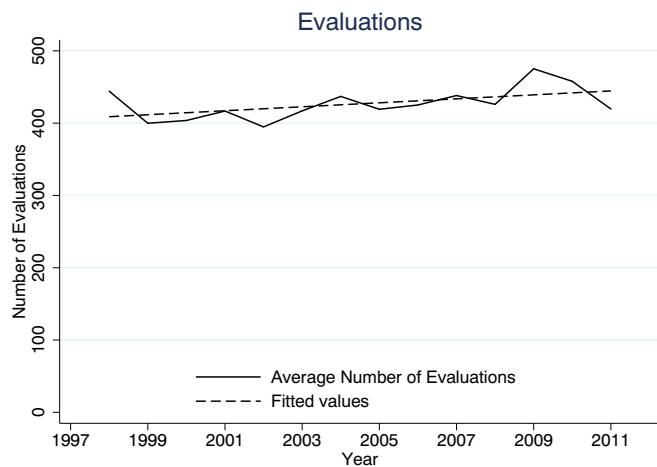


Fig. 4. Average number of evaluations between 1997 and 2011 with estimated trend line.

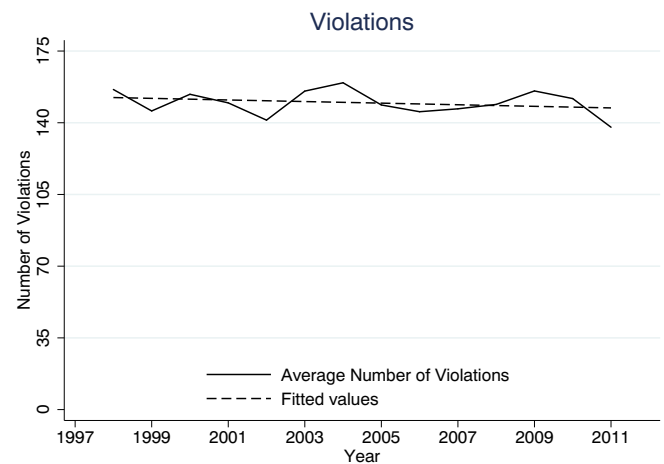


Fig. 5. Average number of violations between 1997 and 2011 with estimated trend line.

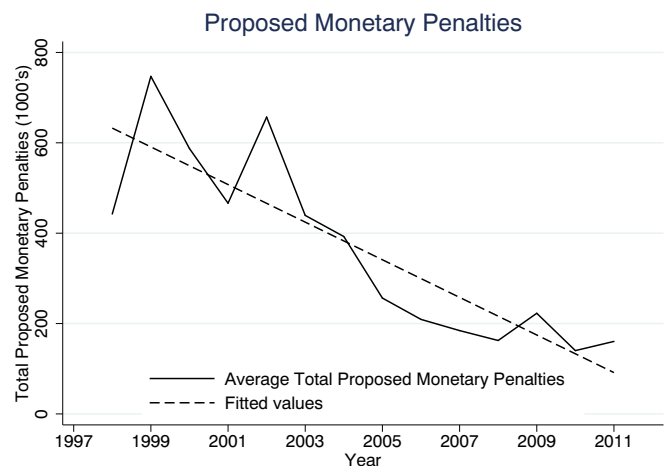


Fig. 6. Average proposed monetary penalty between 1997 and 2011 with estimated trend line.

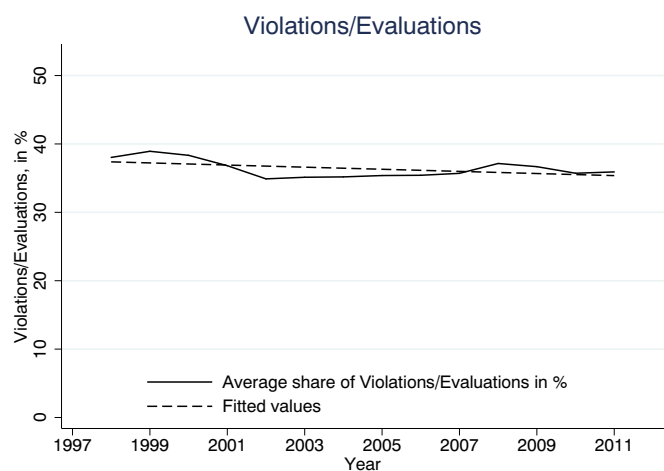


Fig. 7. Average number of violations as share of evaluations between 1997 and 2011 with estimated trend line.

Table 2

Dependent variables include the number of evaluations, violations reported, proposed monetary fine in thousands of dollars, and violations as a share of evaluations in Columns 1–2, 3–4, 5–6, and 7–8, respectively. Standard errors are clustered on the state level. The second column of each outcome variable includes one lag of authorization.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Eval ^{***}	Eval [*]	Vio	Vio	Fine	Fine	Vio/Eval	Vio/Eval
Authorization share	0.620 (0.491)	0.300 (0.571)	0.098 (0.194)	0.065 (0.226)	0.530 (3.615)	1.842 (4.212)	0.0001 (0.0004)	0.0006 (0.0004)
Lagged authorization share		0.607 (0.553)		0.062 (0.219)		−2.478 (4.081)		−0.0010 ^{**} (0.0004)
Observations	700	700	700	700	695	695	700	700
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State time trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors in parentheses.

*** $p < 0.01$.

** $p < 0.05$.

* $p < 0.1$.

but became more lenient in enforcing penalties; or states were better equipped to bring facilities into compliance so that fines were no longer necessary. Next, we ran a regression analysis to unveil any statistical association between the trends.

In Table 2, we use our baseline specification and present the estimation results for the outcome variables (i.e., number of evaluations, number of violations found, total proposed monetary penalties, and violations as a share of evaluations). For each outcome variable, we estimated two regressions: in the first, we only included authorization for the same year; in the second, we included a one-year lag of authorization.

Authorization, whether contemporary or first lag, did not exhibit a significant relationship with either the number of evaluations or the number of violations found. The average proposed monetary penalties showed a decreasing trend over time compared to the increasing trend in authorization, but this relationship was not statistically significant when controlling for covariates and state and year fixed effects. Thus, our null hypothesis of authorization not being related to enforcement outcomes could not be rejected for the number of evaluations, number of violations found, or the fine amount. The point estimates were positive but statistically insignificant. The economic significance was also weak. If we use results from the column including only contemporary authorization values, a one percentage point (pp) increase in authorization would be predicted to increase the number of evaluations by 0.62, the number of reported violations by 0.098, and the value of total fines proposed by \$1842, all of which comprise only a small fraction of the respective variables' mean or standard deviation (see Table 1). If we examine authorization changes across the entire time period, we find the estimated effect to be small as well. The average increase in authorization share was approximately 35 percentage points. Based on the results from Column 1 of Table 2, the total estimated effect on the number of evaluations would then be 0.62×35 , or approximately 22 more annual evaluations. This is equal to roughly 5% of the average total number of evaluations across states and time, which must be considered a rather small effect due to the large absolute and relative increase in our explanatory variable. Hence, statistical and economic significance revealed no relationship between marginal decentralization and enforcement behavior in the context of RCRA program authorization. Our analysis therefore provided no evidence that states assuming more enforcement responsibility notably change their enforcement behavior.

The first lag of authorization indicated a significant relationship with violations as a share of evaluations ($p = 0.019$). The point

estimate implies that a one pp. decrease in authorization increased the violation share by 0.001. This was the only statistically significant relationship between authorization and any outcome variable based on conventional significance levels and may be due to the particular specification rather than a true relationship. The small size of the effect further suggests that the estimated relationship is not meaningful in a real-world sense; the effect is equivalent to a 0.5% decrease of a standard deviation of violation shares.

The results showed no evidence of a race-to-the-bottom scenario. The simplest possible explanation is that the incentives for federal and state authorities are not misaligned when it comes to RCRA enforcement, perhaps because of a lack of serious cross-state externalities. Another explanation is that authorized rules' at-least-as-stringent requirement is working. However, as noted above, while rules should be at least as stringent on paper, it is certainly possible for states to shirk their enforcement responsibilities (Innes and Mitra, 2015; Helland, 1998; Sjöberg, 2016). It should also be noted that firm behavior may change in response to increased decentralization. For example, if firms expect increased leniency in enforcement after state authorization, they might adjust their behavior by decreasing compliance. Hence, to the extent that firms adjust (i.e., increase) environmental risk taking, resultant behavioral modifications could potentially offset the hypothesized leniency effect in state enforcement. Yet, as mentioned in the introduction, there has been very little evidence that decentralization affects environmental outcomes through firm behavior. Also, some studies suggested that firm productivity is not affected by environmental compliance costs (Becker, 2011a), although recent research has further explored this hypothesis (Earnhart and Rassier, 2017).

Our analysis does not provide a clear answer to why states apply for authorization.⁹ If states invest in state-level enforcement infrastructure after decentralization to increase activity through, for example, more evaluations, this scenario is not reflected in our outcome variables. We did not detect any evidence of race-to-the-bottom behavior following from marginal state authorization, and our findings do not imply that state authorization have a positive effect on efficiency in terms of a higher number of evaluations, et cetera. However, a complete study on this topic would require access to data on fiscal resources allocated to environmental enforcement.

In addition, this paper does not explain the reduction in monetary fines over our investigated time period. While we can see that the decrease is not tied to the marginal increase in authorization, we can only

⁹ Scarce research has been conducted to explore why states assume responsibility for enforcing environmental regulations; Stafford (2017) is a recent exception.

speculate about what is behind this trend. Two plausible explanations warranting future research include a change in the type of violations or whether the same type of violation incurs lower penalties over time. The trend could also be driven by changing firm behavior where firms gradually learn which violations are less costly when detected through informal networks or media, et cetera. If penalty levels are more strongly correlated with environmental costs than with firm benefits, firms may avoid more costly violations but continue to commit minor offenses. If the costlier violations constitute only a small share of the total violations, then only the proposed monetary penalties would be affected.

We conducted several robustness checks; results are presented in [Tables A1 and A2](#) in the [Appendix](#). First, we specified the number of evaluations and violations¹⁰ in log form. The authorization share remained insignificant as an explanatory factor; see Columns 1–2 in [Table A1](#) in the [Appendix](#). Second, [Sigman \(2003\)](#) suggested that coastal states act differently from interior states because the latter have stronger freeriding incentives when authorized to enforce CWA rules due to having more neighbors. [Chang et al. \(2014\)](#) also found that coastal states were authorized to enforce CWA rules more quickly than interior states. In an effort to identify any effect in the coastal subset of US states, we interacted authorization with a coastal dummy which is equal to 1 for all coastal states. The results appear in Columns 3–6 in [Table A1](#) in the [Appendix](#). This estimation did not alter our main effect, which remained statistically insignificant for interior states, and the coastal interaction term was not statistically different from the main effect. Third, we further examined interactions between authorization and GDP per capita levels under the hypothesis that richer states might react differently to an increase in authorization compared to poorer states. Specifically, we divided all states into quartiles based on average GDP per capita during our investigated time period and then interacted authorization with dummies for the three highest quartiles using the lowest quartile as the base category. We also investigated the interaction between LCV scores and authorization under the hypothesis that an authorization effect might exist for less environmentally oriented states. None of these interactions showed significant results (see [Table A2](#) in the [Appendix](#)). In sum, all estimated interactions were insignificant, and the point estimates also remained relatively small. As a final robustness check, we ran the regressions removing all years after 2006, when the increasing authorization trend appeared to plateau. The results, presented in Columns 7–10 in [Table A1](#) in the [Appendix](#), indicated no effect on enforcement by any measure, even when authorization share was increasing more rapidly on average.

Finally, we want to briefly discuss our control variables' estimated coefficients. The full set of estimated coefficients is shown for each outcome in [Table A3](#) in the [Appendix](#) using the baseline specification. Not surprisingly, fixed effects explained most of the variance. Neither the GDP nor the LCV score showed a relationship with either of the outcome variables. Democratic states appeared to have somewhat more violations per evaluation than Republican states. The most interesting association is that as the number of firms increased, we observed fewer evaluations and violations. If this association could be explained by reverse causality, it would imply that firms move to states that perform fewer evaluations and find fewer violations. This association suggests an avenue for further research, isolating a potentially causal effect of firms relocating to lenient states. We also noted that the decrease in the number of evaluations associated with an increased number of firms was more substantial than the decrease in the number of violations, such that the predicted violation-to-evaluation ratio rose as the number of firms increased.

4. Summary and Conclusion

State enforcement of federal environmental legislation is a hotly debated topic with no clear scholarly consensus surrounding its effects on environmental or enforcement outcomes. In this paper, we discovered that between 1998 and 2011, more and more states were granted authority over a larger share of RCRA rules. Over these years, the number of evaluations and the number of detected violations remained mostly constant, but the total amount of proposed monetary penalties fell significantly. However, none of these trends showed a statistically significant relationship with the marginal changes in state authorization across states.

Some researchers have expressed concern that decentralization in environmental enforcement may lead to lenient enforcement. On one hand, incentives that lead to a race-to-the bottom may contribute to lax local enforcement; on the other, authorized state programs are at least as stringent or broader in scope than federal regulations ([Levy, 2012](#)). Authorization might also be efficient given states' informational advantage regarding the local environment, local stakeholders' preferences, and local economic and social institutions ([Laskowski et al., 2005](#); [Adler, 2005](#)). In our paper, we found that moving from centralized to at least partially decentralized enforcement represents a policy change that, in the context of the RCRA, is not estimated to affect environmental enforcement. [Oates \(1999\)](#) proposed that concerns about decentralized regulation and interjurisdictional competition lie in the distortion of enforcement and the magnitude of such distortion. Our results showed that any such distortion would be minor, at least with regard to implementation and enforcement levels. It is worth reiterating that during our timeframe, no base programs were authorized; therefore, our results apply to marginal authorization rather than initial authorization. We can conclude that once authorization has been initiated, offering states authority over more RCRA programs does not substantially change enforcement behavior defined as evaluations, violations, and proposed monetary penalties.

[Sigman \(2005\)](#) and [Helland \(1998\)](#) found some evidence of free-riding, and [Grooms \(2015\)](#) identified some effects of decentralization on enforcement outcomes. A major difference between those papers and our study is that the former studied the CWA. This is not necessarily surprising, as different institutional settings can be optimal for the enforcement of different types of environmental legislation given a particular legislation's characteristics. This problem has been discussed in the context of public good provisions (e.g., [Dietz, 2005](#) and [Sproule-Jones, 2005](#)). Using the institutional fit argument in [Dietz \(2005\)](#), [Sproule-Jones \(2005\)](#), and [Epstein et al. \(2015\)](#), water is transboundary in its essence while the cost of poor handling of hazardous waste mostly affects individual states. However, poor handling of hazardous waste may also have transboundary effects, especially when firms are located close to state borders or close to water sources that run across states. Further, the EPA has identified federal oversight of state CWA enforcement as problematic ([EPA, 2011](#)). However, the incentives for states to appear business-friendly, and thus shirk in enforcement, should be the same when implementing the CWA as when implementing the RCRA.¹¹ These scenarios underscore the importance of assigning responsibility to address varied environmental problems to the appropriate institutions. To reconcile the oft-diverging results of the federal enforcement literature, we encourage researchers to pay more attention to the particular characteristics of specific environmental acts. An obvious next step would be to examine whether the data allows a direct comparison of RCRA and CWA enforcement, particularly if the inherent differences in the environmental problems these acts are designed to address may create disparities in enforcement incentives that reveal differences in institutional fit.

¹⁰ Out of 700 observations, six had zero violations. The results were insensitive to whether these observations were omitted or increased by a small value.

¹¹ See Chapter 7 in [Earnhart and Glicksman \(2011\)](#) for a more detailed discussion of state vs. federal enforcement of the CWA.

We have looked at environmental enforcement at the state level; however, local governments may also be important to determining firm behavior through zoning regulations and impact fees that could affect environmental compliance costs (Fischel, 2001). Another promising

direction for future research is to increase the granularity of enforcement based on local government characteristics. This, however, requires a much more detailed dataset than the one to which we have access.

Appendix A

Table A1

Dependent variables include the number of (log) evaluations, (log) violations reported, proposed monetary fine in thousands of dollars, and violations as a share of evaluations in Columns (1), 3, and 7; (2), 4, and 8; 5 and 9; and 6 and 10, respectively. Standard errors are clustered on the state level. The first four columns include an interaction of authorization with a coast dummy, and Columns 4–8 show the results of estimating the base line regression using only years prior to 2007.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Log (evaluations)***	Log (violations)**	Evaluations*	Violations	Fines	Violations/ evaluations	Evaluations	Violations	Fine	Violations/ evaluations
Authorization share	−0.0005 (0.001)	0.0002 (0.001)	0.997 (0.606)	0.291 (0.240)	−0.579 (4.470)	0.0004 (0.0004)	0.834 (0.545)	0.260 (0.193)	2.442 (5.608)	0.0005 (0.0004)
Authorization share * coast			−1.025 (0.965)	−0.526 (0.382)	3.004 (7.112)	−0.0009 (0.0007)				
Observations	700	700	700	700	695	700	450	450	445	450
State fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Authorization lag	No	No	No	No	No	No	No	No	No	No
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State time trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Only years before 2007	No	No	No	No	No	No	Yes	Yes	Yes	Yes

Standard errors in parentheses.

*** $p < 0.01$.

** $p < 0.05$.

* $p < 0.1$.

Table A2

Caption: Dependent variables include the number of evaluations, violations reported, proposed monetary fine in thousands of dollars, and violations as a share of evaluations in Columns 1 and 5, 2 and 6, 3 and 7, and 4 and 8, respectively. Standard errors are clustered on the state level. The first four columns include an interaction of authorization with the LCV, and Columns 4–8 show the results interacting authorization with three dummies representing the three highest quartiles based on average GDP per capita levels between the years 1998–2011.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Evaluations***	Violations**	Fines*	Violations/ evaluations	Evaluations	Violations	Fine	Violations/ evaluations
Authorization share	0.594 (0.894)	0.119 (0.354)	0.464 (6.598)	−0.0006 (0.0007)	1.579 (1.157)	0.219 (0.375)	4.832 (6.855)	0.0001 (0.0001)
Authorization share * LCV	0.001 (0.013)	−0.001 (0.005)	0.001 (0.098)	0.0001 (0.0001)				
Authorization share * p75					−2.282 (1.684)	−0.444 (0.548)	−1.511 (10.021)	−0.0003 (0.0010)
Authorization share * p50					−1.366 (1.285)	0.032 (0.458)	−7.794 (8.286)	0.0003 (0.0009)
Authorization share * p25					1.767 (2.525)	−0.707 (0.742)	−4.974 (13.756)	−0.0011 (0.0014)
Observations	700	700	695	700	450	450	445	450

State fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Authorization lag	No	No	No	No	No	No	No	No
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State time trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors in parentheses.

*** $p < 0.01$.

** $p < 0.05$.

* $p < 0.1$.

Table A3

Dependent variables include the number of evaluations, violations reported, proposed monetary fine in dollars, and violations as a share of evaluations in Columns 1, 2, 3, and 4, respectively. The table shows estimated coefficients for all covariates except fixed effects. Standard errors are clustered on the state level.

Variables	(1)	(3)	(5)	(7)
	Evaluations	Violations	Fines	Violations/evaluations
Authorization share	0.620 (0.491)	0.098 (0.194)	0.530 (3.615)	0.0001 (0.0004)
Log(GDP)	−39.28 (233.43)	−18.68 (92.44)	625.874 (1747.195)	0.164 (0.173)
LCV score	0.013 (0.547)	−0.058 (0.216)	−2.513 (4.071)	0.0004 (0.0004)
Democratic	−25.95* (14.69)	−2.04 (5.82)	−1.965 (108.116)	0.033*** (0.011)
Firms	−0.010*** (0.003)	−0.004*** (0.001)	0.004 (0.023)	0.000004** (0.000002)
Observations	700	700	695	700
State fixed effect	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
Authorization lag	No	No	No	No
Control variables	Yes	Yes	Yes	Yes
State time trends	Yes	Yes	Yes	Yes

Standard errors in parentheses.

*** $p < 0.01$.

** $p < 0.05$.

* $p < 0.1$.

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